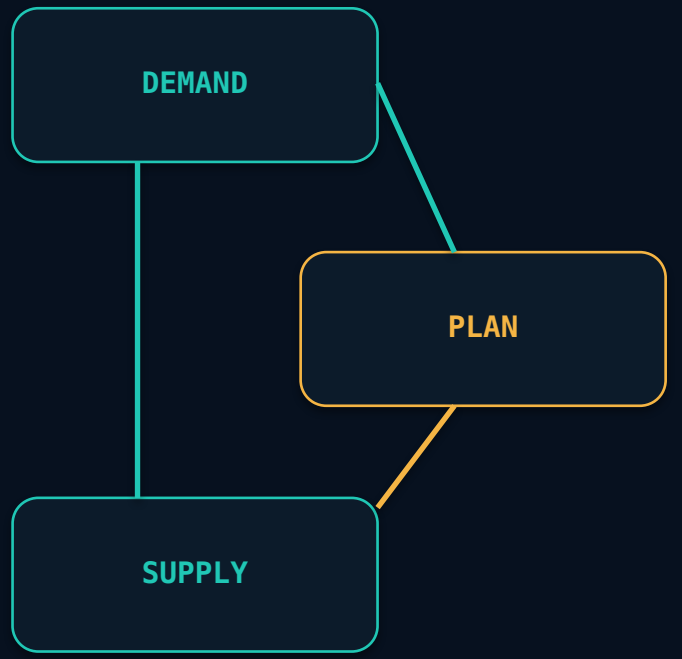
BEGINNER TRAINING

Forecasting methods & seasonality

How demand planners turn uncertain demand into practical supply decisions.



AP1 → AP2

Public-safe learning example • no confidential company data

START HERE

A forecast is a smart estimate, not a promise.

It helps the business prepare before customers ask for a product.

REACTIVE

Without a forecast

The business reacts after shelves are empty, production is late, or cash is tied up in too much stock.

PREPARED

With a forecast

The business orders, produces, staffs, and plans promotion activity earlier.

REALISTIC

But forecasts can be wrong

The job is not to predict perfectly. The job is to reduce surprises and respond faster.

Demand

What customers may buy

Supply

What the business can provide

Plan

The bridge between the two

Demand planning sits between customers and supply.

It converts changing demand signals into a decision about what to buy, make, move, or hold.



Demand planning asks: “What should we expect?” Supply planning asks: “What can we commit to?”

FOUR WORDS TO KNOW

Forecast, actual, error, and bias.

These four ideas are enough to start having a useful planning conversation.

ESTIMATE

Forecast

What we expected to happen. Example: 1,000 units next month.

REALITY

Actual

What really happened. Example: 1,150 units were sold.

GAP

Error

The gap between forecast and actual. Here: 150 units too low.

PATTERN

Bias

A repeated pattern of being too high or too low.

Forecast accuracy is useful. Forecast bias is more important when mistakes repeat.

CORE FORECASTING METHODS

Start simple. Use the method that fits the data and the decision.

Most beginner planning work starts with a baseline, then adds business judgment carefully.

METHOD

Naïve

Next month = last month.
Good for a quick baseline.

METHOD

Moving average

Average several recent months.
Smooths one-off noise.

METHOD

Weighted average

Give recent months more influence.
Useful when demand is changing.

METHOD

Exponential smoothing

A rolling method that updates each month.
Balances history and recent change.

Rule of thumb: always compare a more complex forecast against a simple baseline.

Some demand changes because the calendar changes.

Seasonality is a repeatable pattern by month, week, holiday, weather period, or event cycle.

Example: seasonal apparel

A retailer may sell more cold-weather clothing before winter and more holiday-related items late in the year. The planner should compare the same month across several years, not only the prior month.

Same product. Different months. Different expected demand.

Seasonality does not explain everything. Promotions, competitor moves, stock-outs, and new launches can also change demand.



SEPARATE THE SIGNALS

Trend, seasonality, and one-off events need different responses.

A good plan asks what changed before changing the forecast.

LONGER-TERM DIRECTION

Trend

Demand is gradually moving up or down over time.

Example: a product category is growing each year.

REPEATABLE PATTERN

Seasonality

Demand moves in a familiar calendar pattern.

Example: year-end gifting or weather-related demand.

TEMPORARY DRIVER

One-off event

Demand changes for a specific reason that may not repeat.

Example: promotion, stock-out, launch, or competitor action.

Ask: “Is this a repeating calendar pattern, a lasting trend, or a one-off event?”

AP1 = the demand view of the market.

For this training, AP1 means the initial demand plan before supply constraints are applied. Terminology varies by company.

AP1 asks:

What may customers buy?

It combines the best available demand signals before operations decides what is feasible.

History	Past sales / sell-out
Seasonality	Month, holiday, weather cycle
Commercial input	Promotion, campaign, customer feedback
Channel context	Pipeline and on-hand inventory

AP1 output: a demand expectation, for example 1,375 units.

AP2 = the executable supply decision.

For this training, AP2 means the final plan after inventory, lead time, capacity, and ordering constraints are considered.

AP2 asks:

What can we commit to?

It turns demand intent into an executable supply and replenishment action.

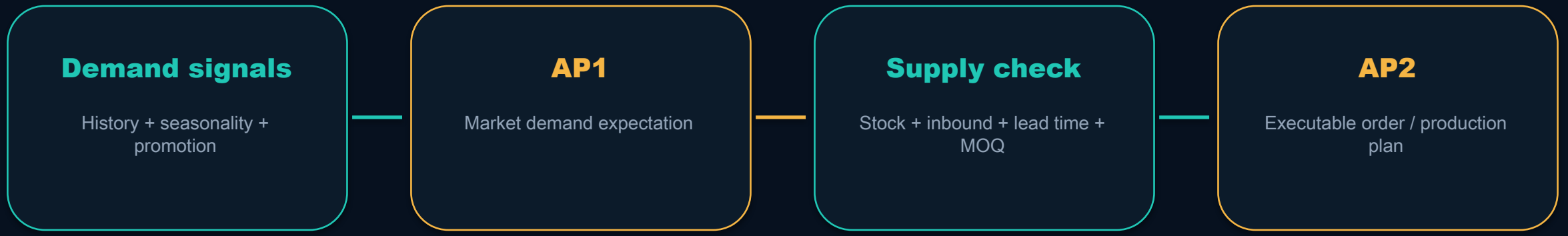
Inventory	What is available now?
Lead time	What arrives in time?
Capacity	What can be made / moved?
Constraints	MOQ, budget, shelf life, allocation

AP2 output: a committed plan, for example 900 units to order now.

AP1 → AP2

The two-stage planning flow.

AP1 protects the customer view. AP2 protects the business from an impossible commitment.



Never skip the supply check: a forecast is not automatically a purchase order.

WORKED EXAMPLE: AP1

A seasonal retail item is expected to sell more next month.

This is an illustrative calculation, not employer data.

HISTORICAL PATTERN

1. Baseline

Average demand for the comparable month: 1,000 units.

CALENDAR EFFECT

2. Seasonal factor

Expected seasonal uplift: +25%.

1,000 × 1.25 = 1,250

COMMERCIAL INPUT

3. Promotion

Planned activity adds +10%.

1,250 × 1.10 = 1,375

OUTPUT

AP1

Demand expectation:

1,375 units

AP1 answers: “If the market is served, how much demand should we be ready for?”

Now check whether the business can actually support the AP1 demand plan.

AP2 converts 1,375 units of expected demand into a supply action.

AP1 demand expectation	**1,375 units**
Inventory available now	450 units
Inbound arriving within lead time	300 units
Safety stock to retain	250 units
Order needed	**1,375 + 250 − (450 + 300) = 875 units**
Supplier MOQ	300 units
AP2 action	**Order 900 units**

AP2 protects service level and avoids ordering a number the supplier cannot accept.

REAL PUBLIC DATA EXAMPLE

Monthly retail sales do not look the same in every month.

Public data helps us see why comparing the right periods matters.

Public retail-data illustration: monthly pattern differs by month



PLANNER TAKEAWAY

How to read it

Do not simply assume December's demand equals January's demand. Use the relevant month, multiple years of history, and known business events to establish the starting point.

Source: FRED series MRTSSM448USN, public U.S. monthly retail-sales data. 2023–2025 monthly averages shown for education.

COMMON BEGINNER MISTAKES

Good demand planning is disciplined thinking, not just a number.

Avoid these traps before you add more complex tools or AI.

FIX: COMPARE LIKE-FOR-LIKE PERIODS

Mistake 1

Using only last month as the answer when demand is seasonal.

FIX: EXPLAIN THE EVENT BEFORE CHANGING THE BASELINE

Mistake 2

Treating a promotion or stock-out as normal demand.

FIX: USE AP2 TO MAKE THE PLAN EXECUTABLE

Mistake 3

Calling AP1 an order quantity without checking supply constraints.

Better planning = transparent assumptions + fast feedback + realistic supply decisions.

RECAP

Forecast demand. Understand seasonality. Make supply executable.

A beginner does not need to memorize every formula. They need to ask the right questions.

1

Forecast method

What does history suggest?

2

Seasonality

What changes because the calendar changes?

3

AP1

What may the market need?

4

AP2

What can the business commit to?

The best next step: pick one product, review its last 24 months, and explain the demand pattern before forecasting it.

Sources: U.S. Census Bureau Monthly Retail Trade; Federal Reserve Bank of St. Louis (FRED), series MRTSSM448USN. AP1/AP2 used here as a teaching convention; terminology varies by company.